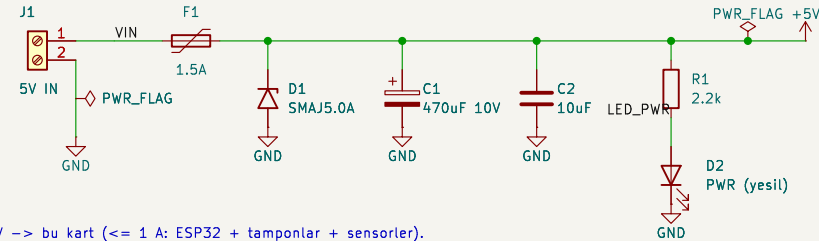
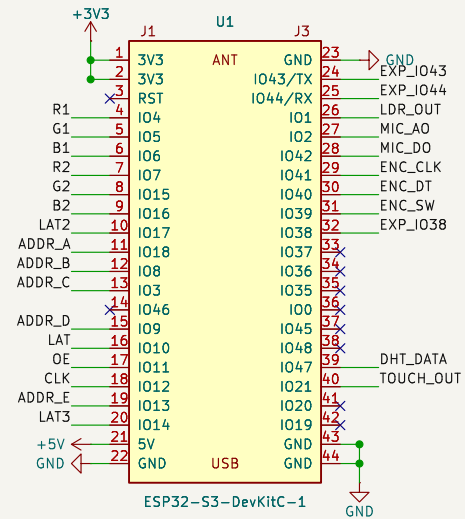


1 5 V GIRIS / KORUMA



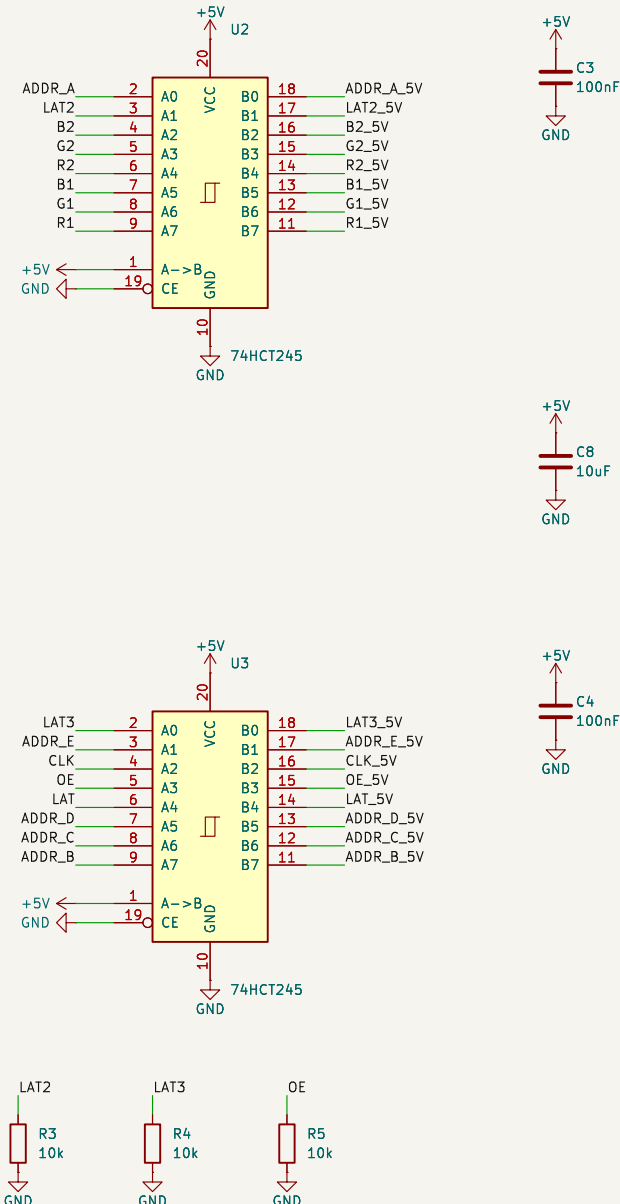
PSU 5 V -> bu kart (<= 1 A: ESP32 + tamponlar + sensorler).
Panel gucu PSU -> panel DOGRUDAN (kalin kablo), bu karttan gecmez.
Ters kutup / asiri gerilim: D1 iletir -> F1 (PTC) acar, kendini sifirlar.

2 ESP32-S3-DevKitC-1 (soket)



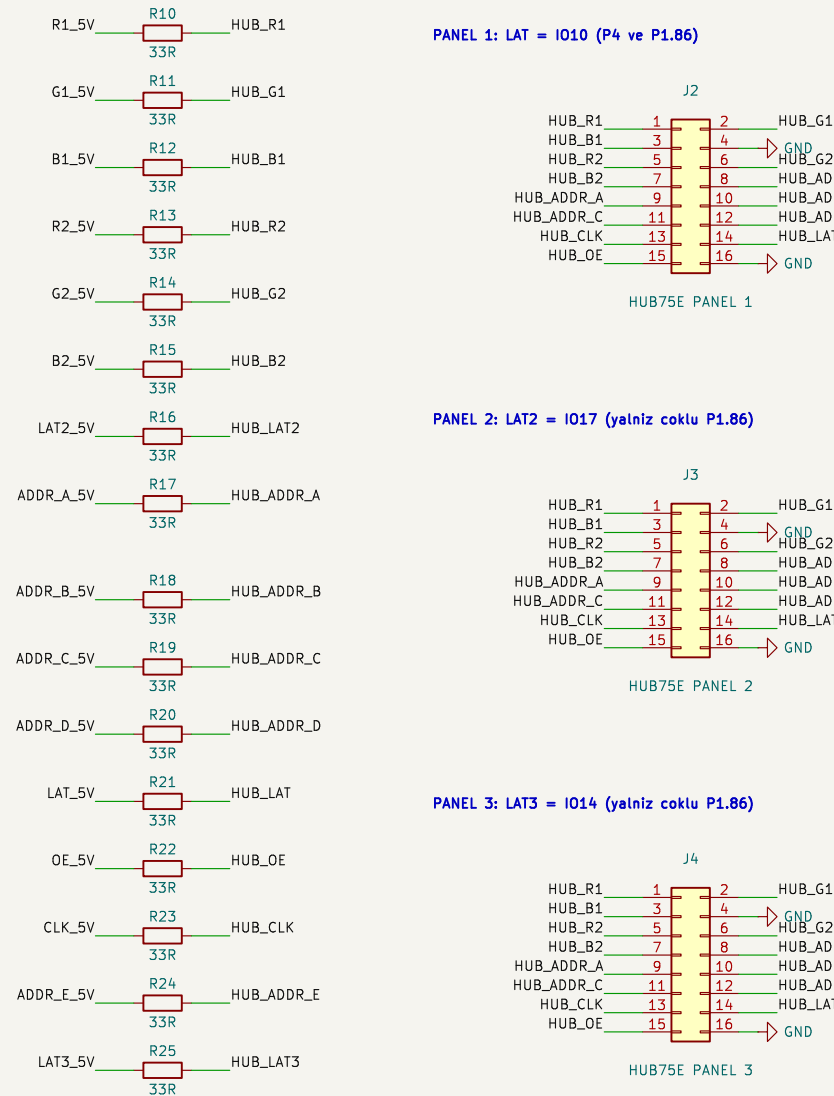
Sag soket 1.0in (25.4 mm; YD-ESP32-S3 tipi N16R8 klon). Resmi kart: 0.9in.
Anten (üst) kart kenarından tasar; USB (alt) kart kenarında.
Kullanılmayan: I00/45/46 strapping, I035-37 OPI PSRAM, I019/20 USB, I048 (klon RGB LED).

3 SEVİYE DONUSTURUCU 3.3 V → 5 V (2x 74HCT245)



Kanal sirasi DevKit J1 pin sirasini izler (PCB: kesisimsiz).
Pull-down (R2-R5): acilista LAT/LAT2/LAT3/OE dusuk -> panel karanlik kalir.

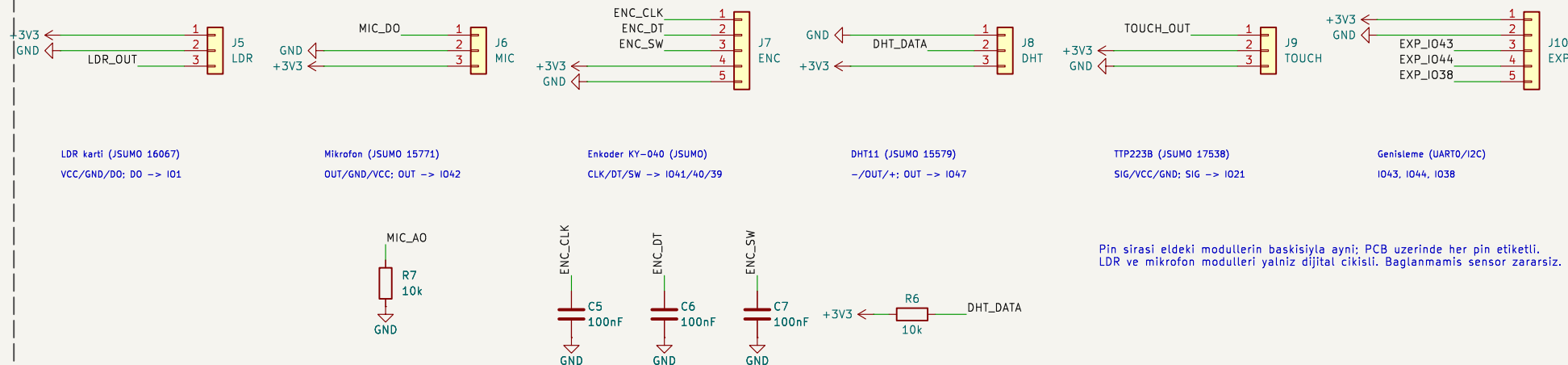
4 HUB75E CIKISLARI (seri 33R, 3 panel)



R10-R25: her hatta 33R seri (0805).
PCB: tek sutun, her direnc kendi
tampon pininin satirinda.

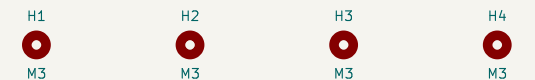
HUB75E: 1 R1 2 G1 3 B1 4 GND 5 R2 6 G2 7 B2 8 E
9 A 10 B 11 C 12 D 13 CLK 14 LAT 15 OE 16 GND
Coklu P1.86: tum hatlar paralel, LAT panel basina ayri.

5 SENSOR / KONTROL BASLIKLARI (3.3 V)



Pin sirasi eldeki modullerin baskisiyla ayni; PCB üzerinde her pin etiketli.
LDR ve mikrofon modulleri yalnız dijital çıkışlı. Bağlanmamış sensor zararsız.

6 MEKANIK



```
hardware/kicad/gen_carrier.py ile uretildi - elle duzenleme yerine ureticiyi degistir
Pinler: include/app_constants.hpp, include/panel_sm16380.h, include/sensors.h
Seviye donusturucu: 2x 74HCT245 DIP (3.3 V -> 5 V), seri 33R, LAT/OE pull-down
MagPanel (github.com/Kem4tk/magpanel)
```

Sheet: /

File: magpanel-carrier.kicad_sch

Title: MagPanel Carrier – ESP32-S3 + 3x HUB75E + sensorler

Size: A3

Date: 2026-10-02

Rev: 1.2

KiCad E.D.A. 8.0.9

1/1